

01211373 Machine Learning and Programming for Industry

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Lecture 00 : Course Introduction

- Course syllabus/grading policy
- Main text & materials
- Introduction to machine learning
- Python & Jupyter notebook guide



Course Syllabus

Name: Machine Learning and Programming for Industry 3(2-3-6)

Description: (from EMME curriculum) Machine learning in industry. Neural network. Hardware and Software. Perceptron. Supervised and unsupervised learning. Reinforcement learning. Loss functions. Back propagation. Binary and multiclass classification. Hyperparameters tuning. Convolutional neural networks. Transfer learning. Data pipelining. Recurrent neural networks. Sequence models. Natural language processing. Embedded device deployment.

Schedule: Thu 1 – 6 PM

Room E5310

Instructor: Dr. Varodom Toochinda

Mobile: 084-3239613 Email: varodom.t@gmail.com

Texts:

- S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.
- Instructor handouts

Grading:

1) Homework Assignments	30 %	2) Project	30 %
3) Midterm	20 %	4) Final	20 %

Google Class Code : nbzzsnrw

<https://classroom.google.com/c/ODU1NjE2Mjc5NDU5?cjc=nbzzsnrw> |



Course Outline

Lecture	Topics
1	ML Foundations & Tooling
2	Classifiers Tour & Data Preprocessing
3	Regression Analysis
4	Ensemble Learning
5	Dimensionality Reduction & Clustering
6	Model Evaluation & Hyperparameter Tuning
7	From Neural Net Foundations to <u>PyTorch</u>
	Midterm
8	<u>PyTorch</u> Mechanics
9	Convolutional <u>neural</u> Networks Convolutional <u>neural</u> Networks
10	Recurrent Networks & Sequential Data
11	Attention & Transformers
12	Reinforcement Learning
13	Deployment & Sim-to-Real
14	Advanced Topics in ML
15	Case study. Course summary
	Final

Main text :

- S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.

<https://sebastianraschka.com/blog/2022/ml-pytorch-book.html>

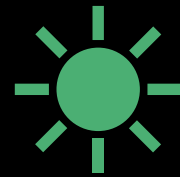


Software



Python with

standard packages
: numpy, scipy,
matplotlib,
panda, etc.
scikit-learn
pytorch



Jupyter
notebook

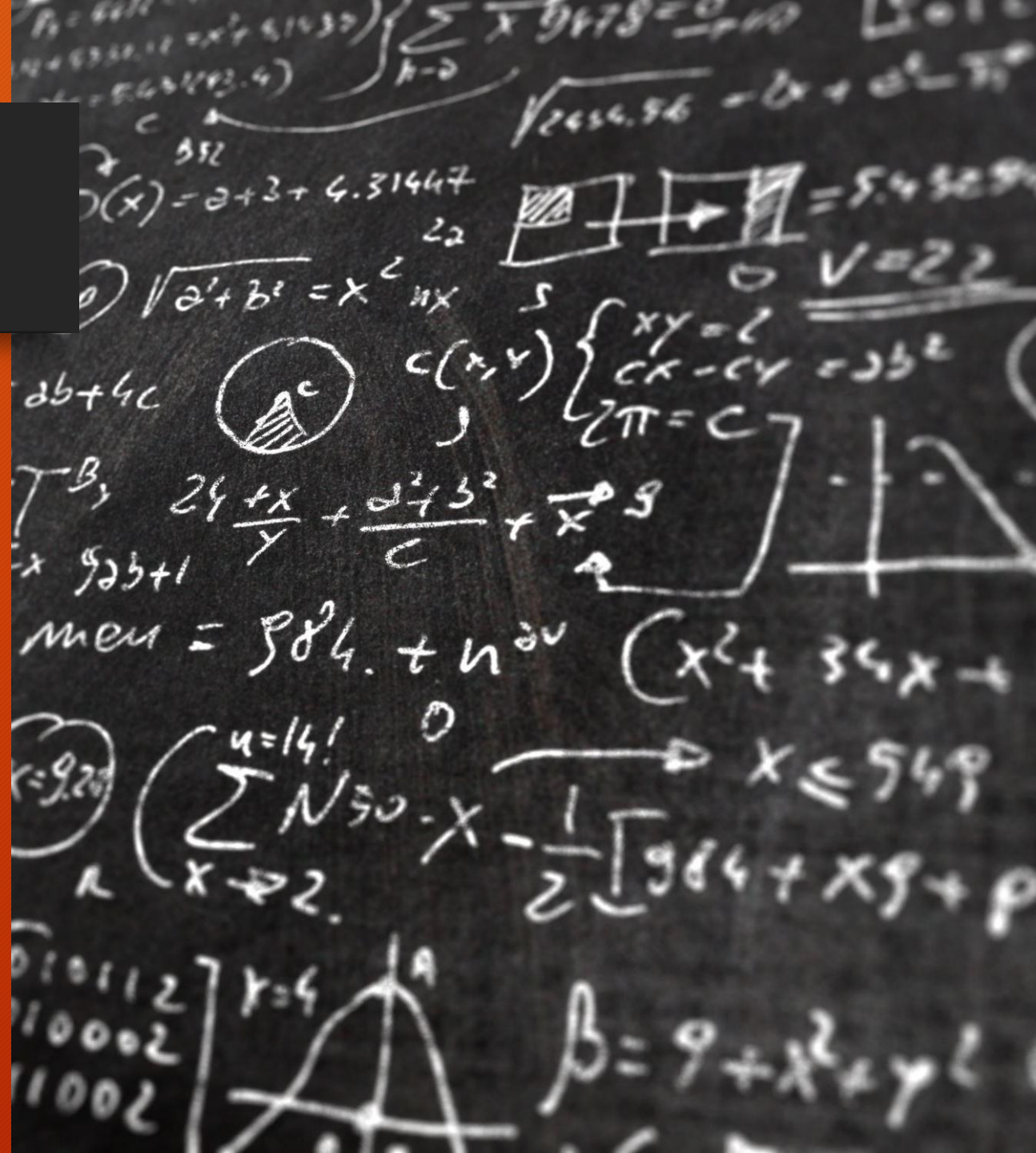
run on Colab
use docker
install locally
(using conda,
venv, uv etc.)



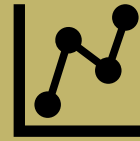
Additional software as
needed

Math

- linear algebra
 - vectors, matrices, tensors
 - eigenvalues/eigenvectors
 - singular value decomposition (SVD)
- calculus
 - derivatives, gradients, jacobians, hessians
- statistics
 - probability, stochastic process



What is machine learning?



Algorithms that analyze data and turn into knowledge

Detect patterns/structures
Classification/regression
Predict future events

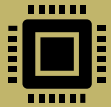


Subclass of AI (Artificial Intelligence)



Trained instead of programmed

Famous quotes

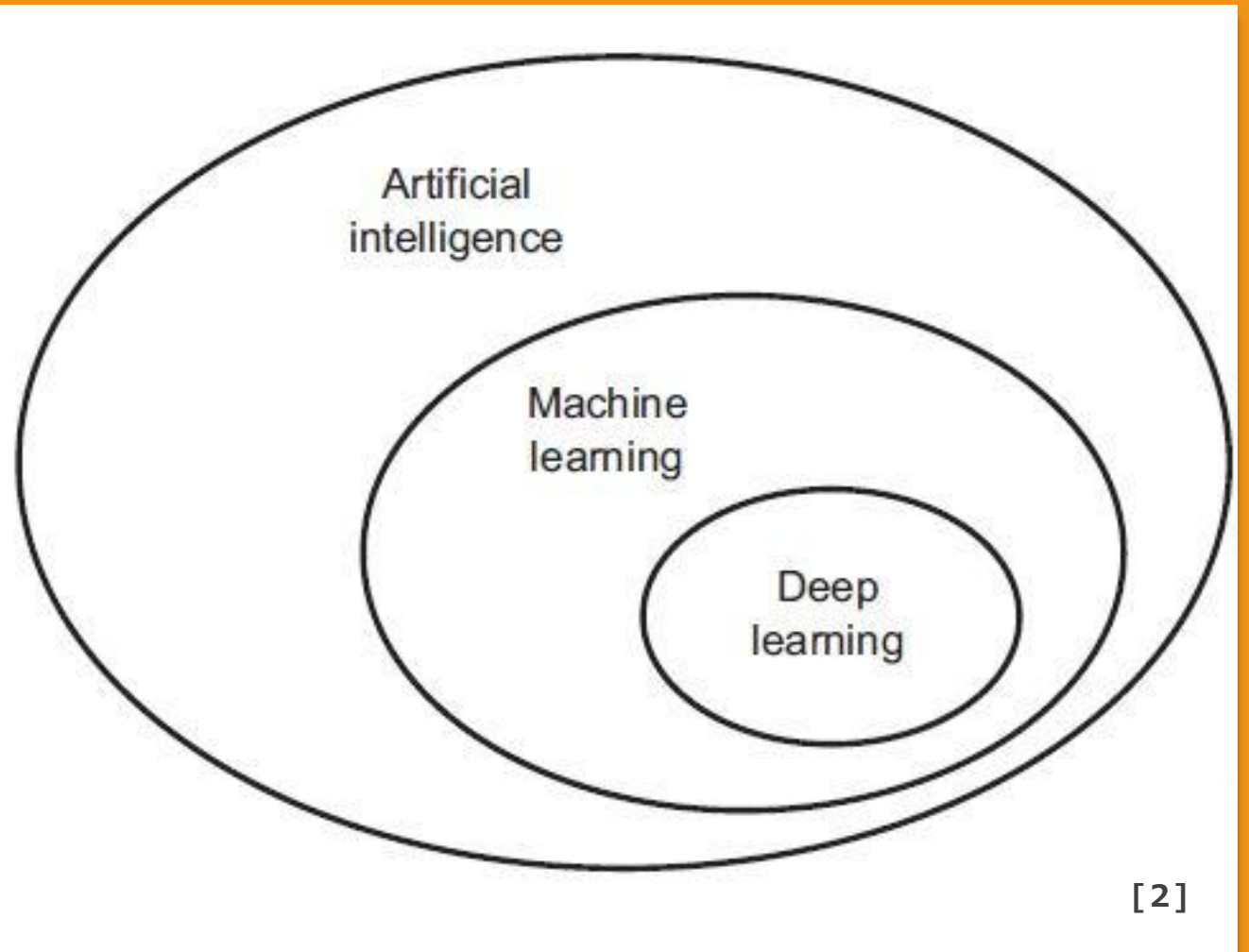


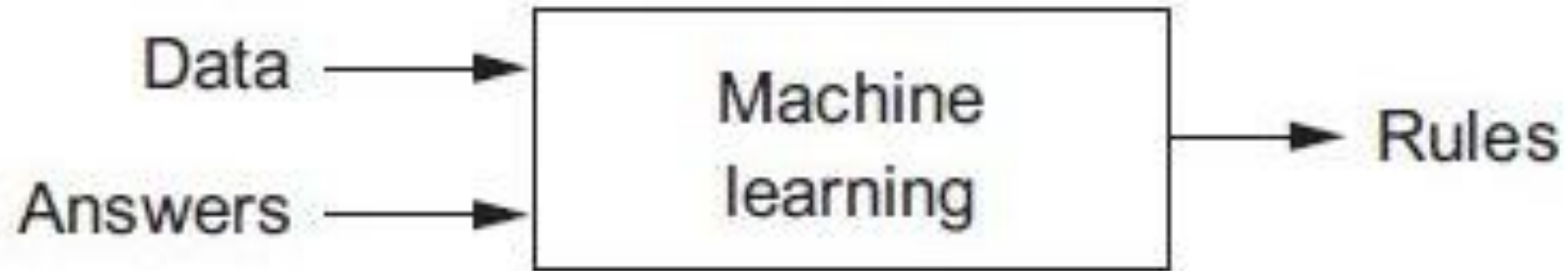
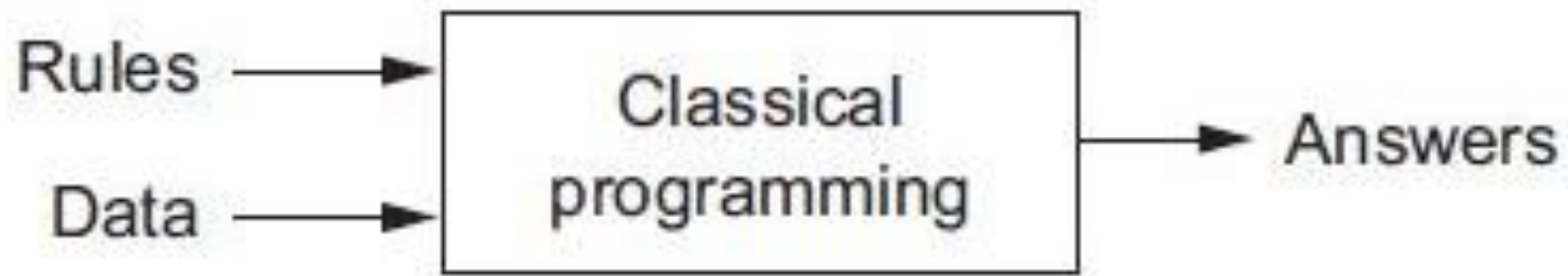
“Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed.” - Arthur L. Samuel . 1959.



“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .” - Tom Mitchell, CMU. 1997.

AI/ML/DL





New program paradigm in ML

[2]

3 types of ML

Supervised

- labeled data (“ground truth”)
- direct feedback
- predict outcome/future

Unsupervised

- no labels/targets
- no feedback
- find hidden structure in data

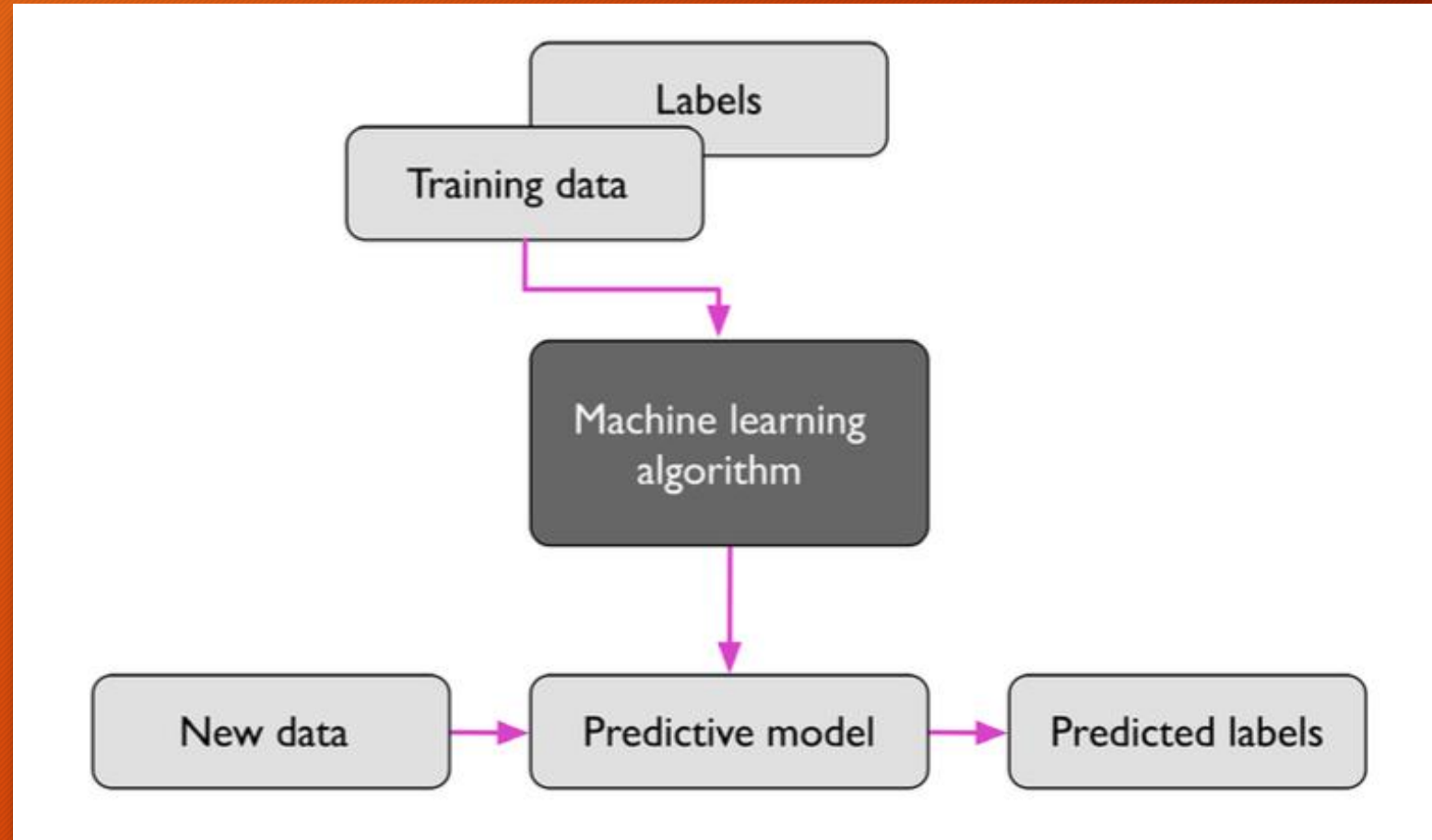
Reinforcement learning (RL)

- decision process
- reward system
- learn series of action

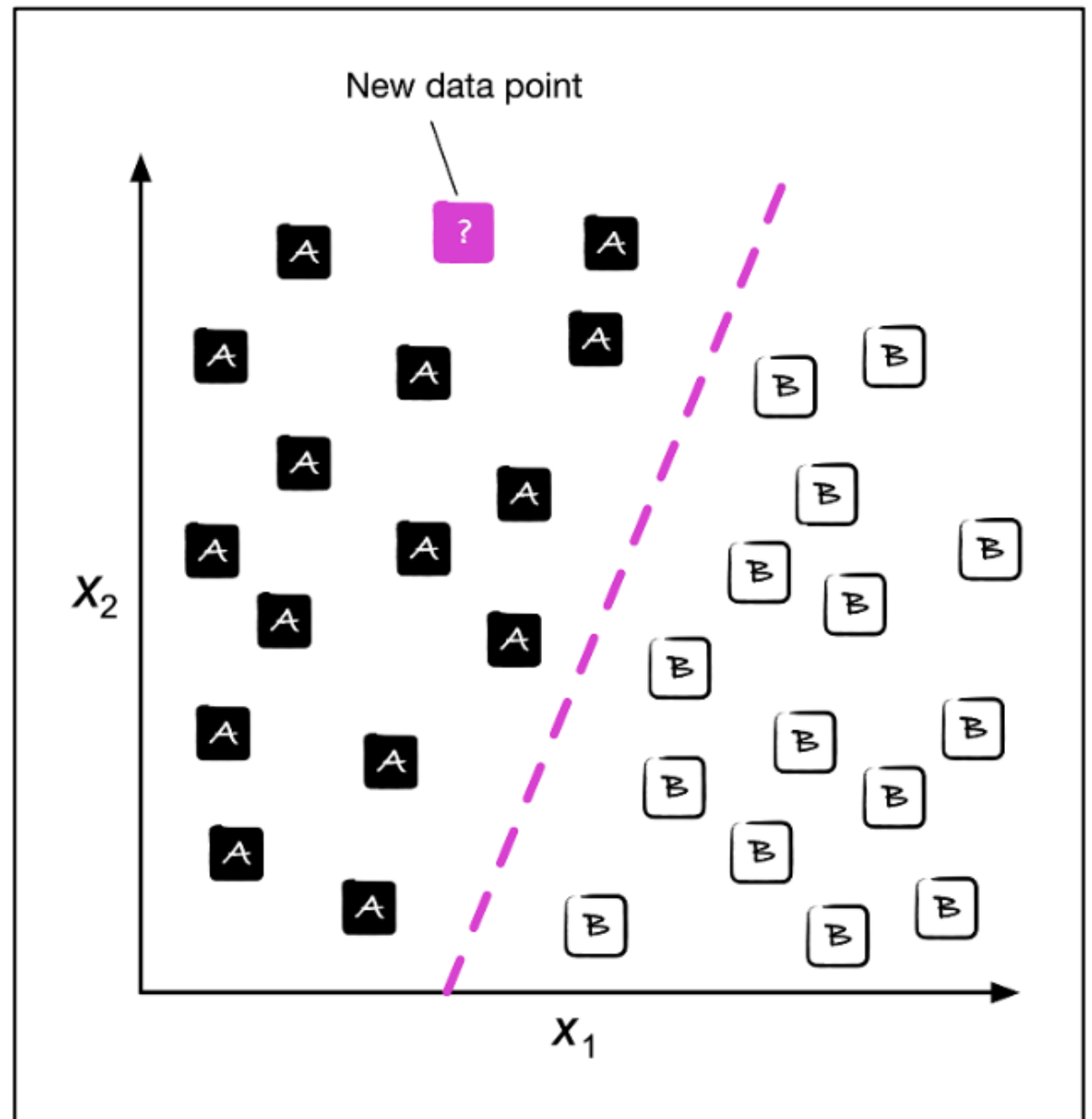
Supervised learning

Supervised learning process

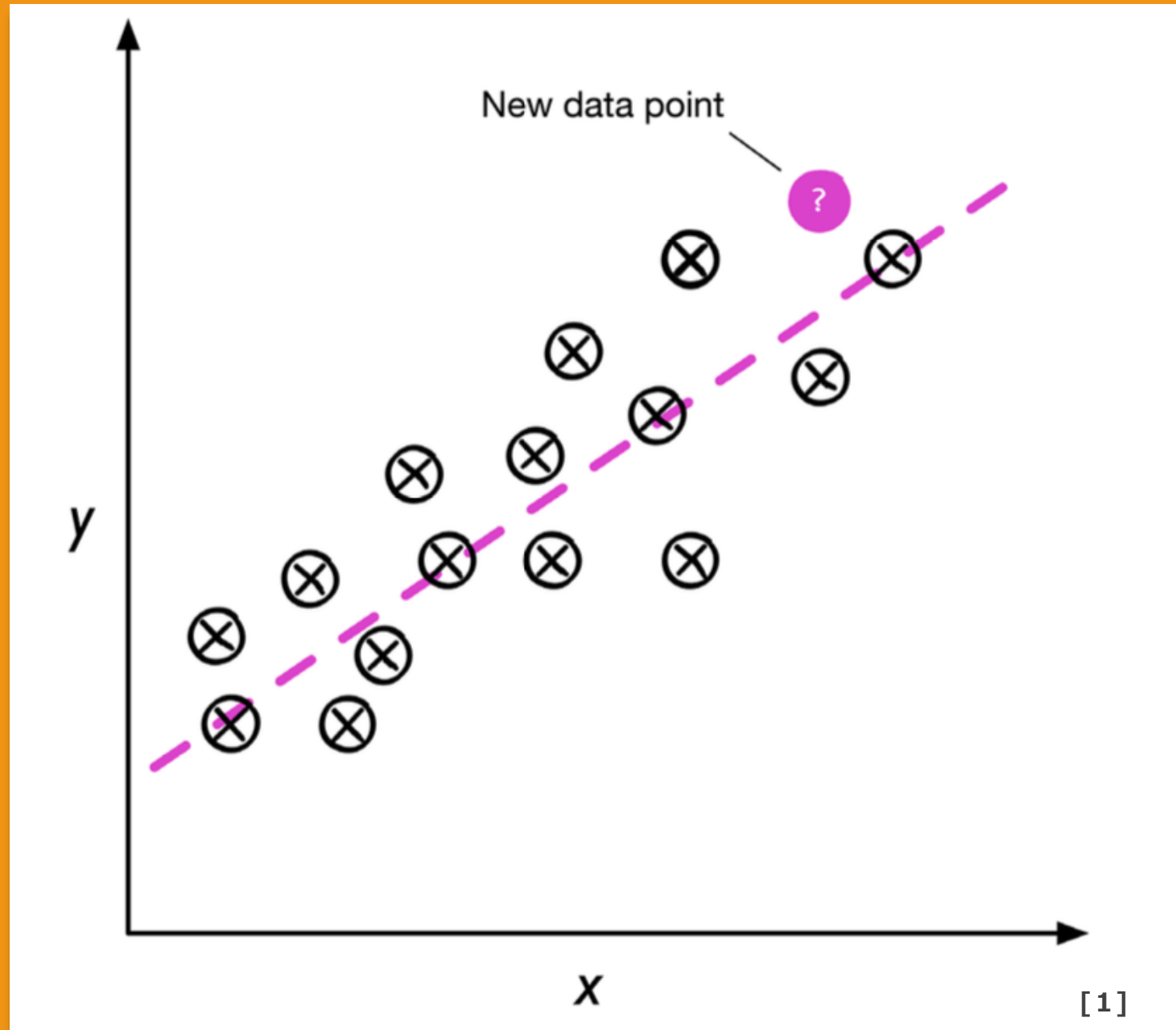
- [1]



Classifying a new data point

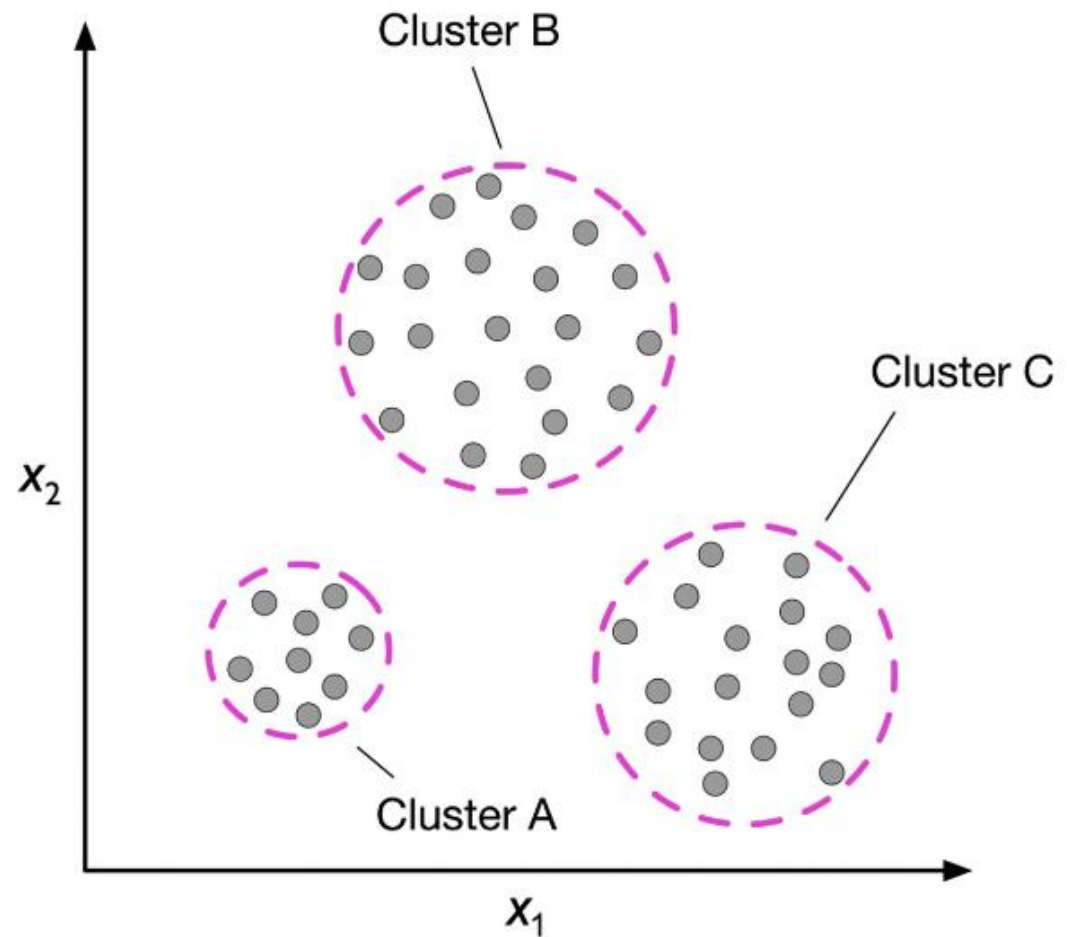


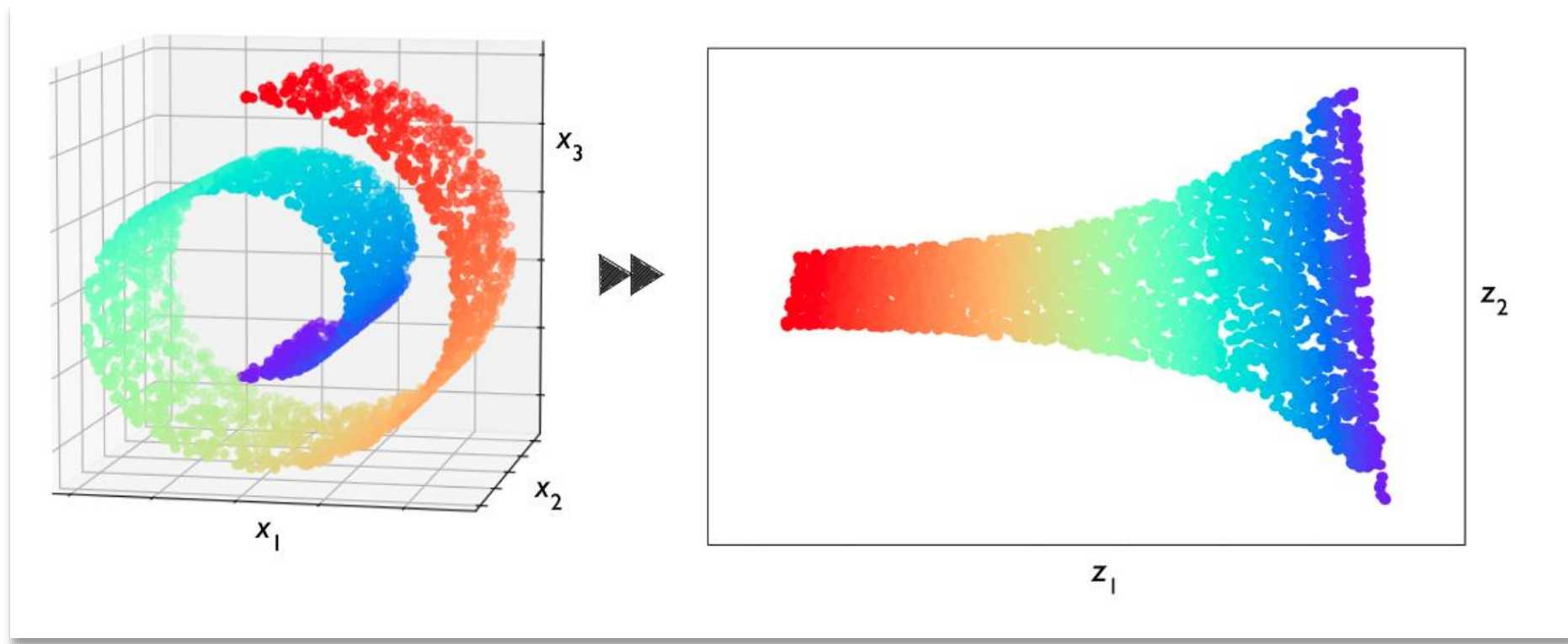
Linear regression



Unsupervised learning

Unsupervised classification : clustering

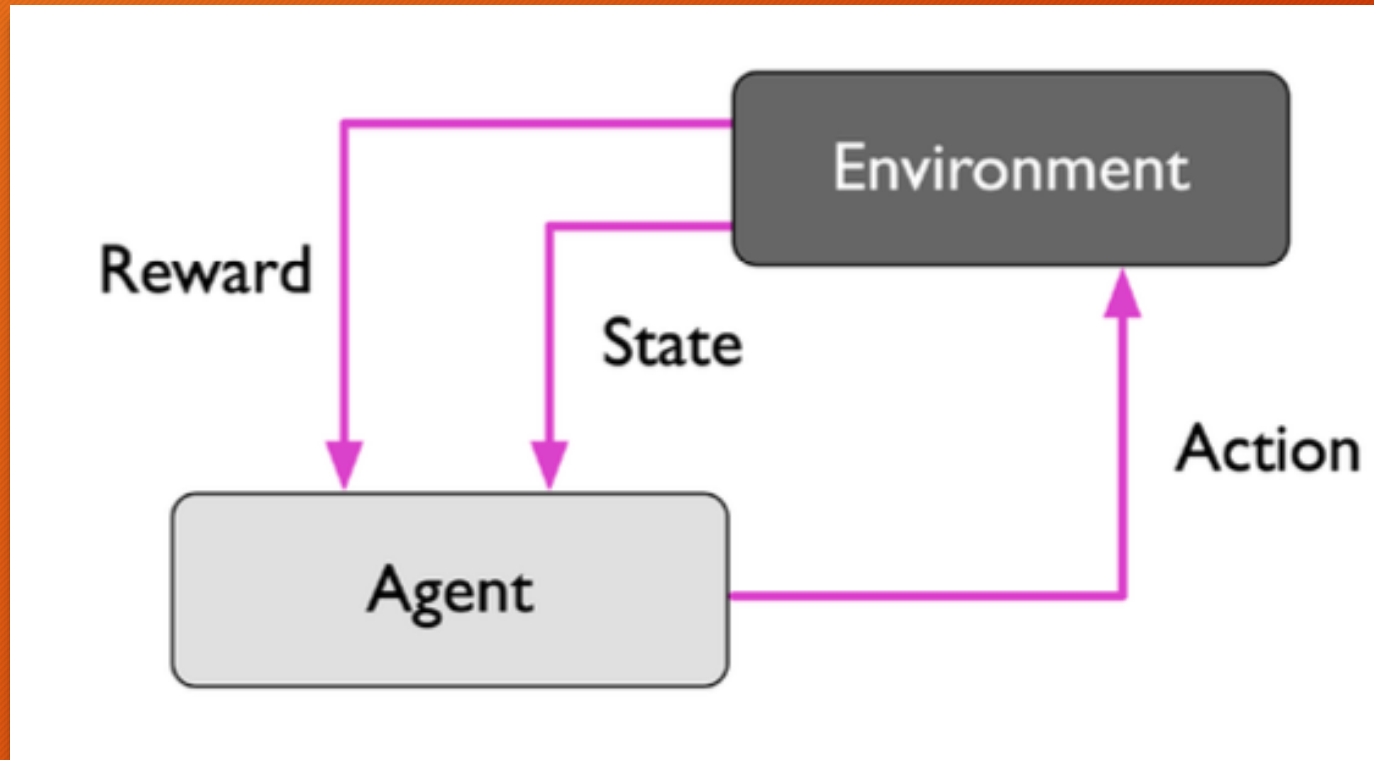




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Dimensionality reduction

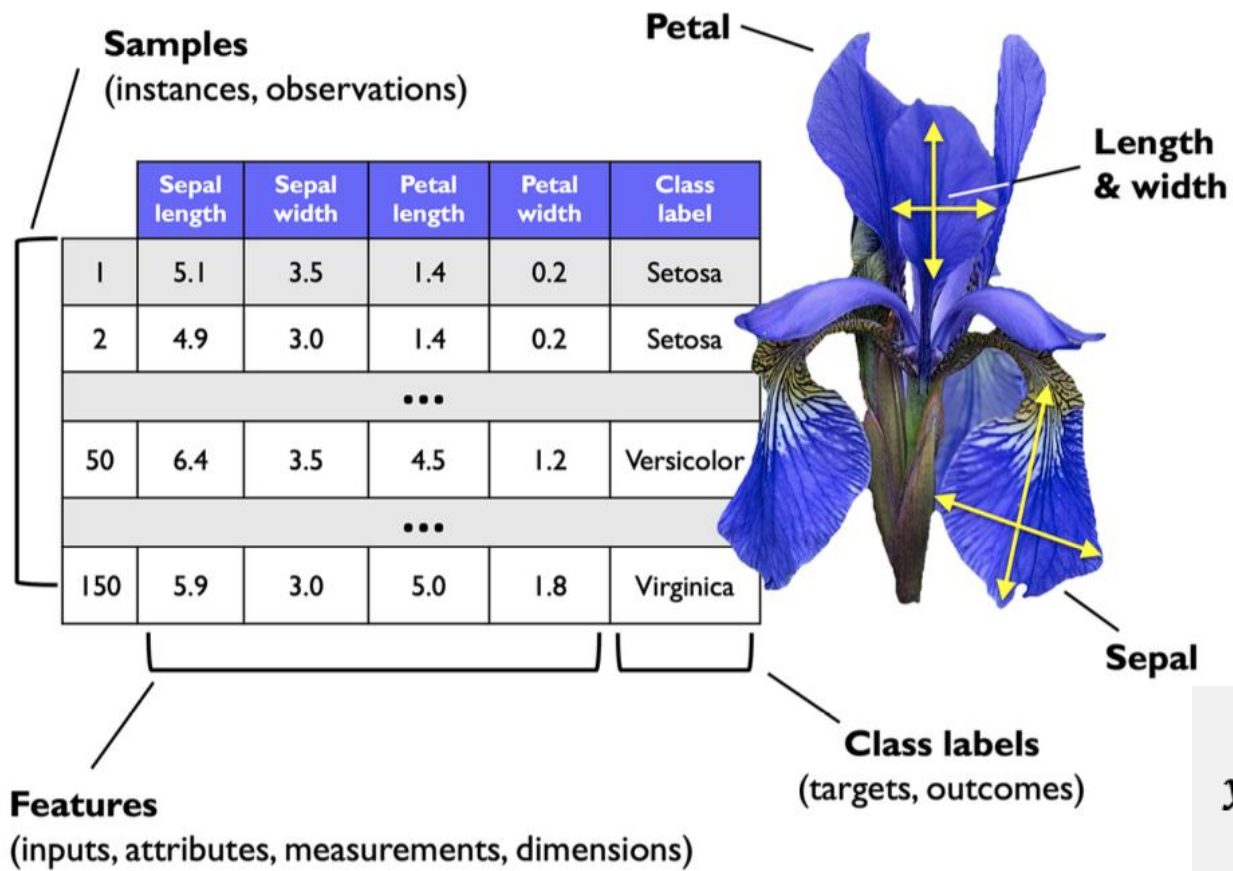
Reinforcement learning



Reinforcement learning process

[1]

Basic terminology and notations



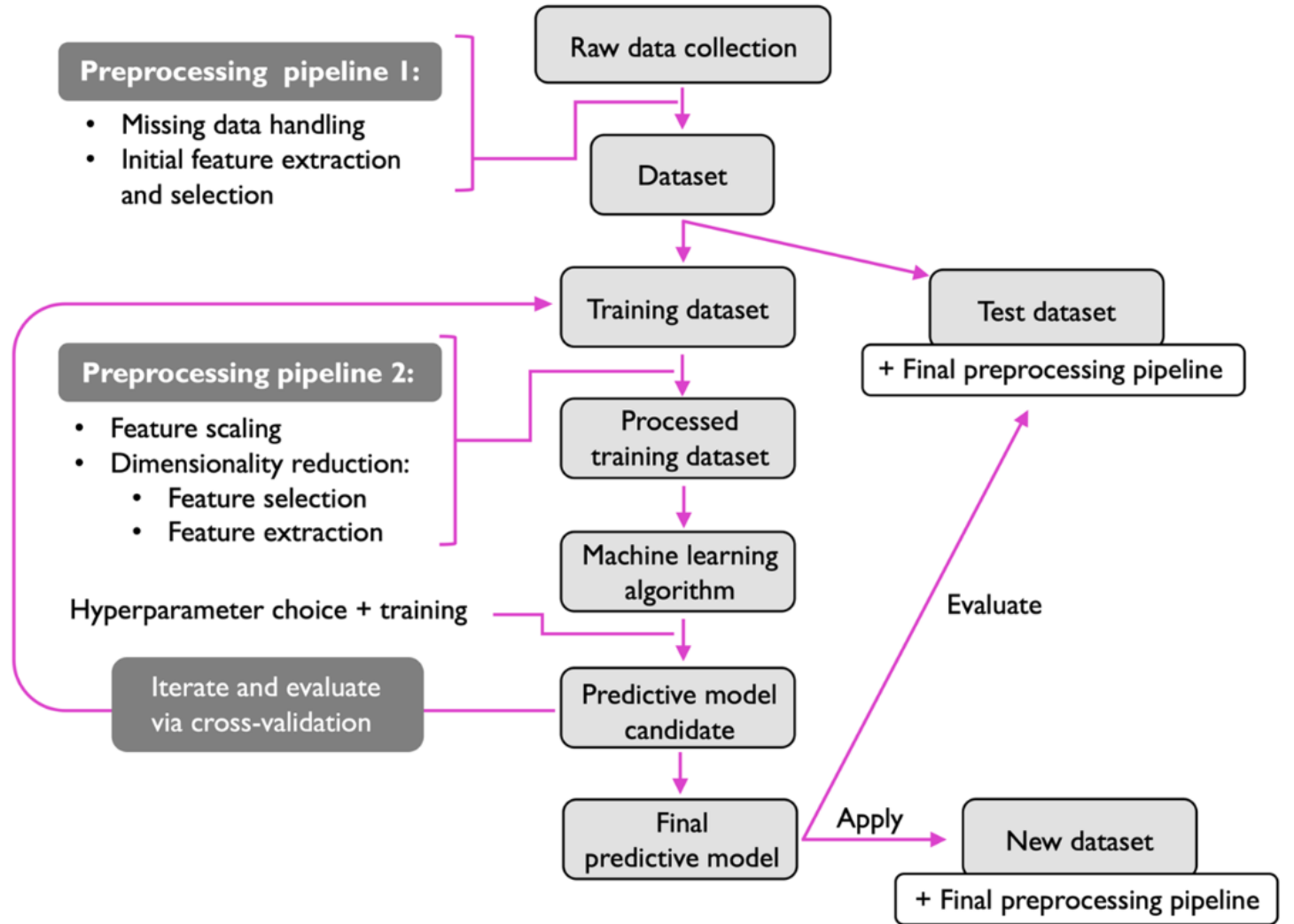
$$X \in \mathbb{R}^{150 \times 4}$$

$$\begin{bmatrix} x_1^{(1)} & x_2^{(1)} & x_3^{(1)} & x_4^{(1)} \\ x_1^{(2)} & x_2^{(2)} & x_3^{(2)} & x_4^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ x_1^{(150)} & x_2^{(150)} & x_3^{(150)} & x_4^{(150)} \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(150)} \end{bmatrix}, \text{ where } y^{(i)} \in \{\text{Setosa, Versicolor, Virginica}\}$$

Example : the iris dataset

Building machine learning system



Problems with previous class



Not tailored to study program
(robotics/industrial automation)



Examples are about flower
classification, house price, used car,
disease diagnostic, etc.

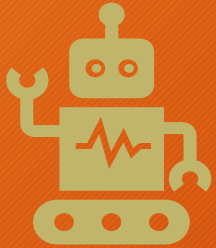


Some methods are less-used
nowadays, especially classical ML.



Newer methods/approaches emerge.

New approach for this course



AI-augmented

Course steered towards robotics/automation

Course materials are mostly AI-generated. So it should be more up-to-date.

AI know which methods are still in use, which are outdated



Challenges

Materials may be new to the instructor as well

Generated labs may have not been tested thoroughly

Setup Python environment locally

Using conda

You should install anaconda or miniconda before executing the following commands on Anaconda prompt.

```
conda install -y jupyter
```

```
conda create -n mlpi
```

```
conda activate mlpi
```

```
conda install conda-forge::jupyterlab
```

```
pip install scikit-learn torch torchvision torchaudio
```

```
python -m ipykernel install --user --name mlpi --display-name "mlpi"
```

Then launch jupyter lab

Using docker

Install docker desktop

Create a work folder and put the provided config.yml and Dockerfile in there

run docker desktop

cd to that folder and type

- docker compose up

Open browser at local:8080

Specify password to launch jupyter notebook

config.yml

```
services:
  notebook:
    build: ./
    ports:
      - 8080:8888
    user: root
    environment:
      - DOCKER_DEFAULT_PLATFORM=linux/amd64
      - JUPYTER_TOKEN=yourpassword
      - NB_UID=1000
      - GRANT_SUDO="yes"
    volumes:
      - ./:/home/jovyan
```

Dockerfile

```
FROM --platform=linux/amd64 quay.io/jupyter/pytorch-notebook
```

```
USER root
```

```
RUN pip install --upgrade pip && \  
    pip install scikit-learn && \  
    fix-permissions "/home/${NB_USER}"
```


- Stanford CS229 (Andrew NG : Autumn 2018)
<https://www.youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShTMGgzqpfVfbU>
- Stanford CS229 (Spring 2022)
https://www.youtube.com/playlist?list=PLoROMvodv4rNyWOpJg_Yh4NSql4Z4vOYy
- MIT 6.036 (Fall 2020)
https://www.youtube.com/playlist?list=PLxC_ffO4q_rW0bqQB80_vcQB09HOA3ClV
- R. Sebastian Intro to ML (Fall 2020)
https://youtube.com/playlist?list=PLTKMiZHVd_2KyGirGEvKlniaWeLOHhUF3&si=CDL0ERiKTeZbKO1U

Some useful
online
courses

References

1. S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.
2. F. Chollet. Deep Learning with Python 2ed. Manning Publications Co. 2021.
3. A. Zhang, Z. Lipton, M. Li and J. Smola. Dive into Deep Learning. 2022.
<https://d2l.ai/>
4. A. Ng. CS229 Lecture Notes. Stanford University. 2023.
https://cs229.stanford.edu/main_notes.pdf
5. T. Broderick. Introduction to Machine Learning course (6.036). Massachusetts Institute of Technology. 2020.
<https://tamarabroderick.com/ml.html>